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GraphRAG-Powered Multi-Agent Research & Learning Intelligence Platform

FINAL YEAR PROJECT 1 - PROPOSAL DEFENCE

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Introduction

What Is RAG?

RAG is an open-book exam for AI. Instead of answering from memory alone, the model looks up relevant material first, then answers using what it just read.

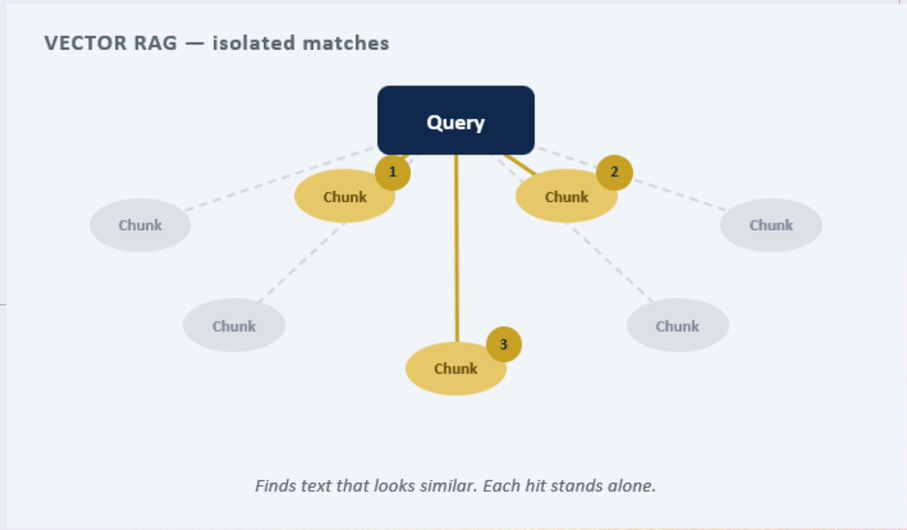
GraphRAG vs Vector RAG

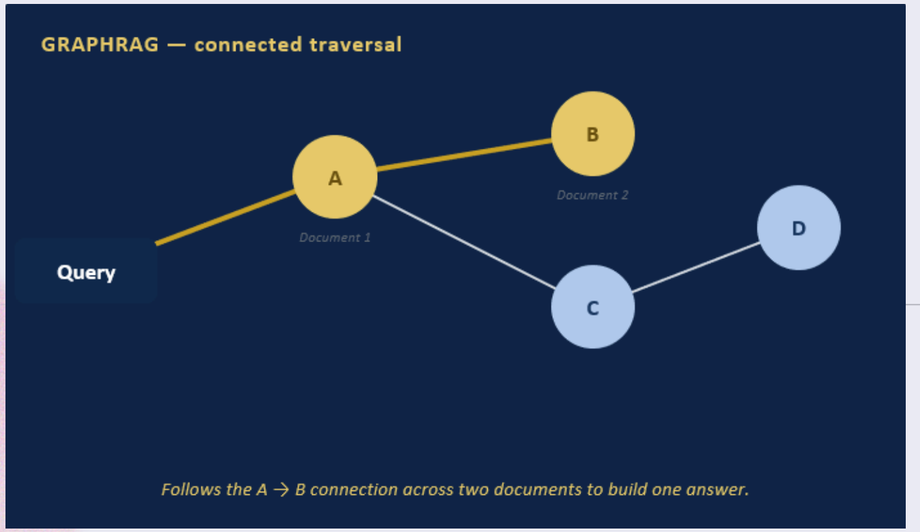
If vector RAG is finding matching pages in a library, GraphRAG is having a librarian who knows how every book connects to every other book and can follow that trail to a complete answer.

Why It Matters

A model trained normally only knows what it memorised during training it can't know about a new paper or a private document.

RAG fixes this because it retrieves relevant text at query time and feeds it to the model as context, so answers are grounded in real sources.





Synthesises across documents automatically

Synthesises across docSurfaces contradictions and connections, not just relevant textuments automatically

Answers corpus-wide questions a single-document RAG never could

Takeaway: Vector RAG finds similar text. GraphRAG finds connected meaning and can answer questions no single chunk could.

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Research Journey

WE ARE HERE — FYP1

START

1 **SIT, May 2025**
Learning and researching AI concepts, RAG, semantic chunking, indexing

2 **SIP, Sept 2025**
Building VERA AI, a production RAG assistant for PETRONAS Upstream BPI Unit

3 **TTP, Jan 2026**
Computer-vision project augmented with a trained model's context to enhance AI output

4 **FYP1, May 2026**
Ingestion, graph construction, hybrid retrieval

5 **FYP2, Sept 2026**
The full graph-native, multi-agent, mode-adaptive platform

6 **Future**
Full production, product enhancement, full STLC, industry-level collaboration

VERA AI: Proof The Limitation Is Real

VERA AI already fuses dense vector search and BM25 keyword matching via Reciprocal Rank Fusion a well-tuned, production-grade hybrid retriever.

But every chunk is still indexed and ranked on its own the system finds the most relevant passage, but cannot tell you that two passages contradict, or that one concept depends on another.

That synthesis is still left entirely to the user. Embedding similarity alone misses relationships between entities across documents, and fails on broad, corpus-wide questions.

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Problem Statement

01

Reactive, Not Predictive

Existing RAG tools store documents as isolated silos with no awareness of cross-document relationships, leaving users to manually connect ideas across sources because the flat storage model cannot do it for them.

02

No Specialised Task Support

Because retrieval never surfaces structured, connected knowledge, there's nothing for a system to route on. A study-plan request, a contradiction check, and a citation request all get the same generic chatbot response.

03

No Personalisation

Because retrieval is flat and undifferentiated, ranking, study-scheduling, and document tagging stay one-size-fits-all — two students uploading the same papers get identical results, regardless of how far along they are.

Underlying cause: Flat, siloed retrieval leaves the system unable to see cross-document relationships so it can't route specialised tasks or personalise to the learner either. Arcana's graph-native retrieval is designed to remove that single point of failure.

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Objective

ARCANA: a graph-native, multi-agent research and learning intelligence platform that turns scattered documents into one living knowledge graph.

01 GraphRAG

Design and implement a graph-native hybrid RAG pipeline extracting entities and relationships from every uploaded document into one interconnected knowledge graph, fusing retrieval across graph traversal, dense vector search, and BM25 via Reciprocal Rank Fusion and demonstrate it outperforms a flat vector-RAG baseline on a quantitative benchmark.

02 GenUI, dashboard & I/O

Build and demonstrate an adaptive 24-component Generative UI covering every input path (document ingestion and chat queries) and every output path (cited summaries, knowledge-graph views, flashcards, and more) streaming through one dashboard that reconfigures itself across Research, Writing, Study, Socratic, and Exploration modes.

03 Agents & tools

Design and implement a composable, LangGraph-orchestrated 25-agent suite each agent equipped with its own tools, communicating only through a single routing function and demonstrate at least one complete end-to-end multi-agent research task across the full pipeline.

04 Trained models

Design the ML Engines behind one shared interface, and train and integrate the Engine as the core deliverable to sit on top of the static heuristics, personalising ranking, spaced repetition, and document handling beyond what fixed rules alone can achieve.

The Single Workspace

The single adaptive workspace holding GraphRAG, the agent suite, the ML engines, and the GenUI layer together for students and researchers reflecting the platform's core mandate of surfacing the hidden cross-document connections that flat, siloed RAG tools leave buried.

The Ecosystem

Operates a modular ecosystem of a 25-agent LangGraph suite, a 24-component Generative UI catalog, and three trained ML Engines, integrated into one platform to execute research, writing, and study tasks.

Name Origin

From the Latin arcana (“secrets, mysteries”) reflecting the platform's core mandate: surfacing the hidden cross-document connections that flat, siloed RAG tools leave buried.

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Literature Review

Domain	#	Key Findings & Arcana Relevance	Gap Addressed
Foundational RAG	2	Gao et al. (2024) maps Naive→Modular RAG; Sarmah et al. (2024) validates hybrid vector + BM25 retrieval	No graph signal in flat hybrid retrieval Arcana adds graph traversal as a third RRF-fused source
GraphRAG	4	Edge et al. (2024) pioneers entity-graph + community summaries; Han et al. (2024) maps the taxonomy; HippoRAG and LightRAG benchmark depth vs. efficiency	Proven for enterprise summarisation only not for an interactive multi-agent research tool
KG Construction	1	Zhang & Soh (2024) propose Extract-Define-Canonicalize for schema-free graph building	No schema-free ingestion pipeline exists for arbitrary student-uploaded documents
Multi-Agent Systems	3	MetaGPT formalises role specialisation; Zheng et al. (2024) validates LangGraph + CrewAI; AdaptOrch (2025) flags the static-topology problem	No LangGraph-based composable agent system purpose-built for research/learning
Adaptive UI	3	ML-identified learning styles improve outcomes; adaptive platforms beat static ones on task completion & satisfaction	Existing tools personalise content only no fully adaptive, mode-switching interface
ML Engines	4	GBDT re-rankers rival neural ones at lower cost; SciRepEval benchmarks doc classification; Half-Life Regression lifts spaced-repetition engagement +12%	No trained re-rank / classifier / mastery layer integrated inside a live hybrid pipeline
Evaluation Frameworks	2	RAGAS gives reference-free faithfulness & relevance scoring; arXiv-corpus caveats inform benchmark design	No RAGAS-style gate comparing graph-hybrid vs. flat-vector retrieval specifically
Generative UI	1	Google's A2UI protocol streams typed UI intent to a trusted component registry	No instantiation as a multi-component, mode-switching research/study/writing catalog

7 Research Gaps

G1: No graph signal in retrieval

G2: GraphRAG unproven for research tools

G3: No schema-free ingestion pipeline

G4: No composable agent system

G5: No mode-switching interface

G6: No trained ML engine layer

G7: No graph-vs-flat eval gate

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Competitive Analysis

Capability	NotebookLM	Elicit	Perplexity	VERA AI (flat RAG)	Arcana
Cross-document graph traversal	×	×	×	×	✓
Multi-hop reasoning	×	Partial	×	×	✓
Contradiction detection	×	×	×	×	✓
Specialised multi-agent routing	×	×	×	×	✓
Adaptive UI per task mode	×	×	×	×	✓
Trained ML personalisation	×	×	×	×	✓
Grounded citations	✓	✓	Partial	✓	✓
Works on private/local corpus	✓	×	×	✓	✓

Aspect	Vector RAG	GraphRAG
Retrieval unit	Isolated text chunks, ranked by embedding similarity	Entities + relationships graph nodes and edges
Reasoning	Single-hop one similarity match per query	Multi-hop traverses connected entities across documents
Corpus-wide questions	Fails on "what are the main themes?" style queries	Pre-built community summaries answer global queries directly
Cross-document links	Each passage encoded and retrieved in isolation	Knowledge integrated across passage and document boundaries
Compute trade-off	Cheaper to index; less relational depth	Higher indexing cost, recoverable via leaner designs (LightRAG)

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Methodology & Work Plan

FYP 1 PROPOSAL DEFENCE

Agile, Iterative Development

P0 → P1 → P2

RAGAS-Gated Empirical Evaluation

P0 **Weeks 1 – 9**

Ingestion pipeline, knowledge-graph construction, hybrid retrieval (dense vector + BM25 + graph traversal via RRF), and the first three agents online end-to-end.

P0 **Weeks 10 – 20**

24 agents across 4 tiers, 24 GenUI components, the UIBlock streaming protocol, all 5 task modes, and the 3-panel adaptive dashboard.

P0 **Weeks 21 – 30**

Re-Rank Engine training, RAGAS evaluation harness, the graph-hybrid vs. flat-RAG benchmark, the structured user study, and dissertation writing.

~30 Weeks (FYP1 + FYP2)

Research & Architecture	GraphRAG & Retrieval & I/O	Full Agent Suite & GenUI	ML engine Traning Evaluation & Benchmarking	Refinement Reporting & Submission
• Lit review (20 papers, 7 domains) + problem scoping • Tech stack, repo, architecture design, GraphStore abstraction	• Ingestion pipeline + KG construction (NetworkX) • Hybrid retrieval (vector + BM25 + graph, RRF); 3 agents live • Smoke-tested end-to-end; interim report	• All agents / 4 tiers, all GenUI components, UIBlock protocol • All 5 task modes, 3-panel dashboard, Firebase Auth/Firestore • Per-agent + full-flow validated; docs + tech paper draft	• Re-Rank Engine training + Neo4j migration • RAGAS harness + benchmark dataset • Graph-hybrid vs. flat-RAG benchmark; user study	• Final fixes, export/interop, system hardening • Dissertation: draft (W9) → tech paper (W10) → hard-bound (W14) • Final validation, rehearsal, presentation (W12)

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System Architecture

FYP 1 PROPOSAL DEFENCE

A DOCUMENT INGESTION PIPELINE

Data Injestion Pipeline

Uploaded Docs PDF · DOCX · APIs → **Upload & Parse** multi-format → **Chunk & Embed** 3072-dim vectors → **Extract Entities** & relations · schema-free EDC → **Document Classifier** tags each chunk → **Simultaneous Write** tags each chunk

Unified Data Tier

- **Primary Store** Firestore / Firebase
- **Vector DB** Pinecone
- **Knowledge Graph** Neo4j · NetworkX

B AGENTIC REASONING PIPELINE

P0 core
P1 primary
P2 extended

User Input entry point → *MultiState AgentState* → **LangGraph Orchestrator** Orchestrator Agent · P0

Multi-State Firestore · stateful routing

Single Routing Function

AGENT SUITE composable · phase-tiered · 25 agents

Core Agent
Orchestrator · Research · Graph · Ingestion

Support Agent
Learning · Writing · Analytics · Memory · Discovery · UI Agent · Citation · Document · Annotation · Study Planner · Fact Checker · Web Search · Socratic Tutor · Visual

Export/Utility Agent
Methodology · Debate · Audio · Plagiarism · Grammar & Style · Paraphrase · Export

Retrevial & Context fusion

Session Memory (Short Term/Long Term)

Web Search (tool call)

C ADAPTIVE GENERATIVE UI

Session Memory (Short Term/Long Term) → **Selects component** UI Agent · P1

TYPED UIBLOCK CATALOG rendered across all task modes

CitedSummary · SourceList · ContradictionAlert · GapAnalysis · InsightCard · KnowledgeGraphView · ConceptMap · ComparisonChart · Timeline · DataTable · FlashcardDeck · QuizCard · BlurtingPrompt · FeynmanExplainer · CornellNotes · SocraticDialog · StudyPlanner · DraftEditor · AudioSummary · PlagiarismReport · BibliographyExport · LiteratureMatrix · ProgressDashboard · CitationPreview

Adaptive Dashboard
reshapes per task mode
Research · Study · Writing · Socratic · Exploration

Multimodal Export
PDF Report, Word Document, Obsidian Vault, Literature Review, BibTex, Slide deck

Emit UIBlock
typed · SSE stream

Hybrid Retrieval
Graph + Vector + BM25 · fused via RRF → **Hybrid Retrieval Engine**

Shared Interface

Session Memory (Short Term/Long Term)

D PERSONALIZED ML ENGINES

- **Re-Rank Engine**
- **Mastery Engine**
- **Document Classifier**

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Tech Stack

ML Training & Data

TensorFlow
Re-Rank / Mastery / Doc Classifier training

Google Colab
GPU training environment

Hugging Face
Pretrained model hub

Kaggle
Public training datasets

RETRIEVAL & KNOWLEDGE STORES

Pinecone
3,072-dim vectors

OpenAI Embeddings
text-embedding-3-large

Neo4j / NetworkX
GraphStore abstraction

Hybrid Fusion
Graph + Vector + BM25 via RRF

BACKEND & ORCHESTRATION

FastAPI
Python 3.11, async-native API layer

Python 3.11
Core backend runtime

LangGraph
Stateful execution graph, 25-agent suite

LLM & Generative APIs

Claude API
Primary LLM

OpenRouter
Fallback LLM routing

OpenAI Embeddings
text-embedding-3-large - vector generation

Gemini 3.1 Flash Image
Image generation - Nano Banana 2

DATABASE & AUTHENTICATION

Firestore
Metadata & user profiles

Firebase
Hosting & Cloud Functions

Firebase Auth
Email / Google OAuth

FRONTEND & GENUI

Next.js 14
App Router · Hybrid RSC/Client rendering

Tailwind CSS
Centralised theme

Vercel AI SDK
Typed GenUI component streaming

Work Plan Gantt Chart

FYP1 - Weeks -1, 1–14

No.	Detail / Week	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Selection of Project Topic															
2	Preliminary Research Work															
3	Submission of Progress Assessment 1															
4	Proposal Defence															
5	Submission of Interim Draft Report															
6	Submission of Progress Assessment 2															
7	Submission of Interim Report															

FYP2 — Weeks 1–15

No.	Detail / Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Project Work Continues															
2	Submission of Progress Assessment 1															
3	Submission of Draft Dissertation and Draft Technical Paper															
4	Submission of Dissertation (soft bound) and Revised Technical Paper															
5	Project Presentation															
6	Submission of Progress Assessment 2															
7	Submission of Project Dissertation (Hard Bound)															

Process Suggested Milestone

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Expected Outcome & Conclusion

OBJECTIVES STATUS

01	**Knowledge & Retrieval**	Done	Pipeline built; quantitative benchmark vs. flat-RAG baseline still to run
02	**Agents & Tools**	Done	Composable suite built, end-to-end multi-agent task demonstrated
03	**Trained ML Models**	In Progress	Training pipeline built; engines not yet trained
04	**Generative UI, Dashboard & I/O**	In Progress	Component catalog and adaptive dashboard semi built but not integrated into the app

WHAT ARCANA DELIVERS

- **Graph-native retrieval :** A graph-hybrid pipeline (vector + BM25 + graph traversal via RRF) replaces flat document silos, enabling cross-document queries flat vector-RAG can't answer.
- **Composable agent suite :** A LangGraph-orchestrated agent pipeline replaces a single general-purpose chatbot, each agent specialised for research, writing, study, Socratic, or exploration tasks.
- **Adaptive Generative UI :** A mode-adaptive component catalog replaces a static chat interface, reshaping layout and tools per task mode in real time.

CONTRIBUTIONS

- **Academic :** A reproducible comparison of graph-hybrid vs. flat vector-RAG retrieval, contributing evidence on whether graph traversal measurably improves multi-hop research queries, building on the VERA AI baseline.
- **Practical :** A working, demonstrable multi-agent research and learning platform exercised end-to-end across all adaptive task modes on real documents.

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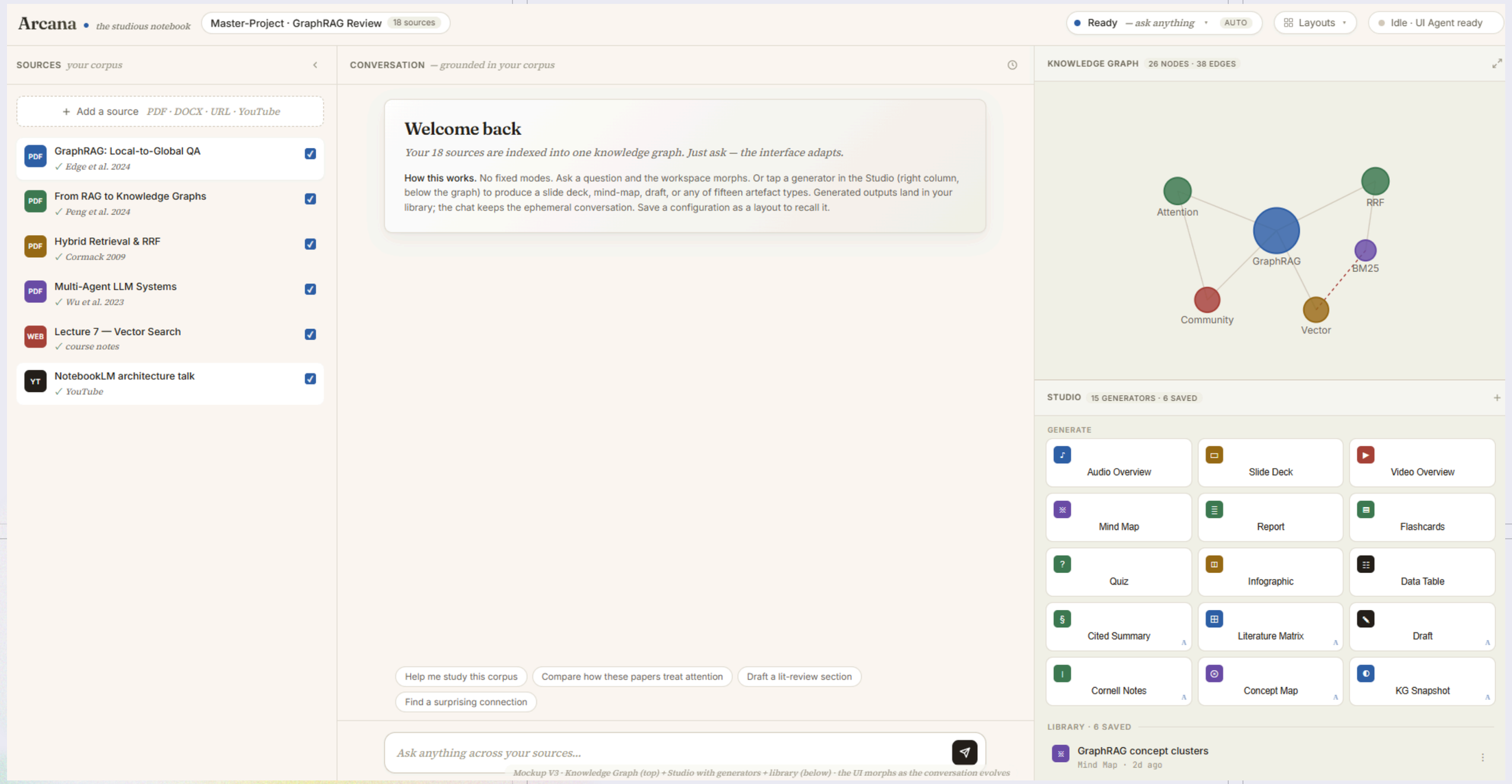
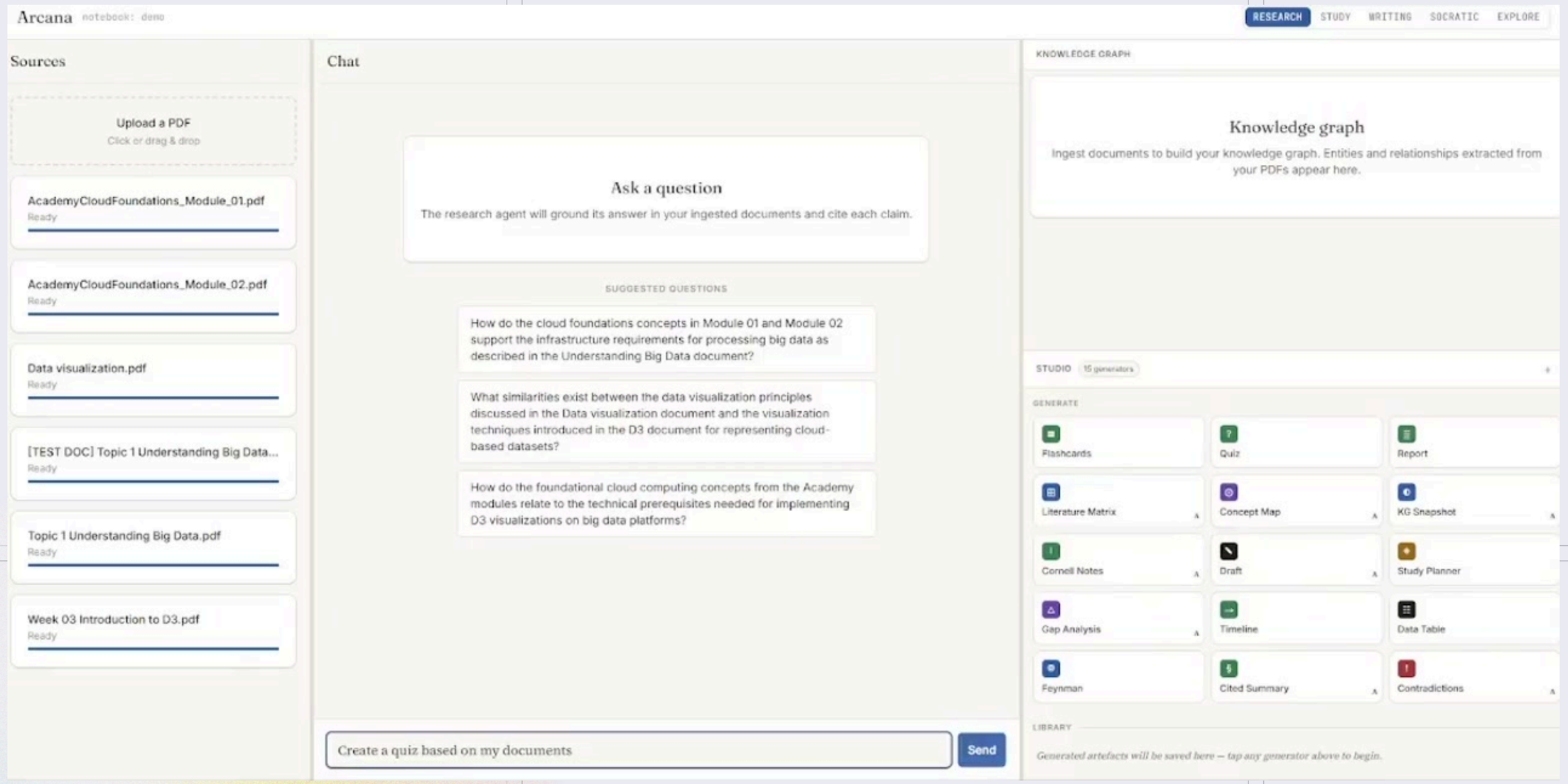
Dashboard Demo

FYP 1 PROPOSAL DEFENCE



Live Demo


Arcana
notebook: demo
RESEARCH
STUDY
WRITING
SOCRATIC
EXPLORE
Sources
Upload a PDF
Click or drag & drop
AcademyCloudFoundations_Module_01.pdf
Ready
AcademyCloudFoundations_Module_02.pdf
Ready
Data visualization.pdf
Ready
[TEST DOC] Topic 1 Understanding Big Data...
Ready
Topic 1 Understanding Big Data.pdf
Ready
Week 03 Introduction to D3.pdf
Ready
Chat
Ask a question
The research agent will ground its answer in your ingested documents and cite each claim.
SUGGESTED QUESTIONS
How do the cloud foundations concepts in Module 01 and Module 02 support the infrastructure requirements for processing big data as described in the Understanding Big Data document?
What similarities exist between the data visualization principles discussed in the Data visualization document and the visualization techniques introduced in the D3 document for representing cloud-based datasets?
How do the foundational cloud computing concepts from the Academy modules relate to the technical prerequisites needed for implementing D3 visualizations on big data platforms?
Create a quiz based on my documents
Send
KNOWLEDGE GRAPH
Knowledge graph
Ingest documents to build your knowledge graph. Entities and relationships extracted from your PDFs appear here.
STUDIO
15 generators
GENERATE
Flashcards
Quiz
Report
Literature Matrix
Concept Map
KG Snapshot
Cornell Notes
Draft
Study Planner
Gap Analysis
Timeline
Data Table
Feynman
Cited Summary
Contradictions
LIBRARY
Generated artefacts will be saved here — tap any generator above to begin.

Future Work

Evaluation 1 - Retrieval Benchmark

QUANTITATIVE

CLAIM

Graph-hybrid retrieval (vector + BM25 + graph traversal, RRF) outperforms flat vector-RAG on cross-document queries

BASELINE

Flat vector-RAG same embedding model, same document set, no graph traversal

DATASET

Held-out, multi-hop cross-document question set size finalised with supervisor

METRICS

RAGAS gate: faithfulness, answer relevance, context precision, context recall

STATUS

Author-run smoke benchmarks complete; full RAGAS-gated comparison scheduled for the next build phase

Evaluation 2 - User Study

QUALITATIVE + TASK-BASED

CLAIM

The adaptive GenUI experience is coherent and usable across all adaptive task modes

METHOD

Structured tasks assigned per mode (Research, Study, Writing, Socratic, Exploration)

PARTICIPANTS

Representative student sample; recruitment finalised with supervisor

METRICS

Task completion rate, mode-switch friction, perceived usefulness per mode

STATUS

Study design complete; recruitment scheduled with supervisor for the next build phase

References

Foundational RAG

NO	Authors & Year	Key Findings	Relevance	Gap Addressed
1	Gao et al., 2024, Foundational RAG	Canonical RAG survey; maps Naive -> Advanced -> Modular RAG progression.	Establishes why plain RAG is insufficient and motivates the move to GraphRAG.	Doesn't add a graph signal; Arcana extends the taxonomy with graph traversal.
2	Sarmah et al., 2024	Validates hybrid dense-vector + BM25 retrieval over either method alone.	Direct mid-baseline between plain vector RAG and Arcana's full graph-hybrid pipeline.	Stops at vector+BM25; Arcana adds graph traversal as a third fused signal (RRF).
3	Zhang & Soh, 2024	Extract-Define-Canonicalize (EDC): schema-free entity/relationship extraction, no pre-defined ontology required.	Principled method for Arcana's ingestion pipeline, where uploaded documents have no fixed schema.	A general extraction method; Arcana operationalises it for arbitrary, student-uploaded research documents.

GraphRAG

NO	Authors & Year	Key Findings	Relevance	Gap Addressed
4	Edge et al., 2024	Builds an entity KG from documents, pre-generates community summaries for query-focused summarisation.	Primary foundational citation; the paper that coined the GraphRAG paradigm.	Targets enterprise summarisation; Arcana extends it to an interactive multi-agent research/learning tool.
5	Han et al., 2024	First comprehensive survey of GraphRAG: indexing, graph-guided retrieval, graph-enhanced generation.	Justifies where Arcana's hybrid retrieval (graph + vector + BM25) sits in the broader taxonomy.	Maps the landscape but proposes no system; Arcana implements a concrete point in that space.
6	Gutierrez et al., 2024	HippoRAG: schema-free KG + Personalised PageRank, modelled on hippocampal memory indexing.	Closest existing system to Arcana's knowledge-graph layer; a natural comparison point.	No composable multi-agent layer or adaptive UI; Arcana adds both on top of a similar graph core.
7	Guo et al., 2024	LightRAG: dual-level (low/high) retrieval, comparable accuracy to GraphRAG at a fraction of the token cost.	Informs the NetworkX-vs-Neo4j efficiency/depth trade-off behind Arcana's GraphStore abstraction.	An efficiency baseline only; doesn't address multi-agent orchestration or UI adaptivity.

References

Multi-Agent

NO	Authors & Year	Key Findings	Relevance	Gap Addressed
8	**Zhuge et al., 2024**	MetaGPT: role-specialised multi-agent collaboration with defined roles, tools, and shared workflow.	Theoretical basis for Arcana's role-specialised agent suite (Research, Writing, Learning, etc.).	Generic software-engineering agents; Arcana applies role-specialisation to research and learning tasks.
9	**Zheng et al., 2024**	Evaluates LangGraph + CrewAI in combination for multi-agent task allocation and information transmission.	Validates LangGraph as an orchestration framework, the exact tool Arcana is built on.	Evaluates the framework pairing in general; doesn't apply it to a research/learning assistant.
10	**Tran et al., 2025**	AdaptOrch: models task decomposition as DAGs to solve the 'topology selection problem' in static LangGraph pipelines.	Identifies the exact limitation of static LangGraph graphs that Arcana's composable, route_to_agent design responds to.	Proposes automated topology selection; Arcana's hop/token-budgeted routing is a complementary, deployable answer.

Adaptive UI

NO	Authors & Year	Key Findings	Relevance	Gap Addressed
11	**Essa et al., 2023**	Systematic review: ML-identified learning styles improve personalised adaptive learning outcomes.	Academic grounding for Arcana's adaptive Learning/Study mode.	Reviews the premise only; Arcana operationalises it as a mode-switching Generative UI.
12	**Dascalu & Harris, 2025**	Mixed-methods UX study (23 participants) comparing adaptive learning platforms on completion time, satisfaction, accuracy.	Closest methodological template for Arcana's planned user study (similar pool size, comparable outcome metrics).	Compares existing platforms; doesn't evaluate a graph-native, multi-agent research assistant.
13	**Strielkowski et al., 2025**	Bibliometric review: adaptive-learning publications grew from 1 (1990) to 636 (2023).	Macro-level evidence that personalised adaptive AI learning is a rapidly scaling, timely field.	Establishes field momentum only; no system contribution to compare against.
14	**Google A2UI Team, 2025**	Declarative, framework-agnostic protocol for agents to stream typed UI intent to a trusted component registry.	Arcana's typed UIBlock{type,data,meta} SSE-streamed protocol is explicitly modelled in the spirit of A2UI.	A general cross-platform spec; Arcana instantiates it as a 24-component, 5-mode research/study/writing catalog.

References

Machine Learning

NO	Authors & Year	Key Findings	Relevance	Gap Addressed
15	**Qin et al., 2021**	Gradient-boosted decision trees remain competitive with, or superior to, neural rankers on many benchmarks.	Justifies the Re-Rank Engine's GBDT design over a neural re-ranker.	Compares ranker families in isolation; Arcana applies the finding inside a live, RRF-fused retrieval step.
16	**Sun et al., 2023**	LLMs can act as zero-shot re-ranking agents, but at high inference cost vs lightweight learned rankers.	Frames the cost/accuracy trade-off Arcana resolves with a small trained GBDT Re-Rank Engine instead of an LLM re-ranker.	Studies LLM re-ranking alone, not as a swappable, optional post-RRF step in a larger pipeline.
17	**Singh et al., 2023**	SciRepEval: multi-format benchmark for scientific document representation and classification.	Grounds the design of Arcana's Document Classifier Engine over existing chunk embeddings.	Provides a benchmark, not a deployed classifier; Arcana integrates the idea into a live multi-agent pipeline.
18	**Settles & Meeder, 2016**	Half-Life Regression: trainable spaced-repetition model; +12% engagement in a live Duolingo study.	Direct basis for the Mastery Engine's design (HLR, swappable against SM-2).	Applied to vocabulary review only; Arcana applies HLR to flashcards generated from arbitrary uploaded documents.

Evaluation

NO	Authors & Year	Key Findings	Relevance	Gap Addressed
19	**Es, James et al., 2023**	RAGAS: reference-free, automated framework for evaluating RAG faithfulness, relevance, and context precision.	Direct basis for Arcana's RAGAS evaluation gate comparing graph-hybrid retrieval vs the flat-vector baseline.	Evaluates generic RAG; Arcana applies it specifically to a graph-vs-flat retrieval benchmark.
20	**Clement et al., 2019**	Characterises arXiv's suitability and caveats as a research corpus for NLP/ML experiments.	Informs caveats for constructing Arcana's own literature-style retrieval benchmark corpus.	A generic dataset-methodology paper; Arcana applies its cautions to a student-facing benchmark specifically.

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THANK YOU

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QNA